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Performance Analysis on Complementary FET (CFET) Relative to Standard CMOS With Nanosheet FET

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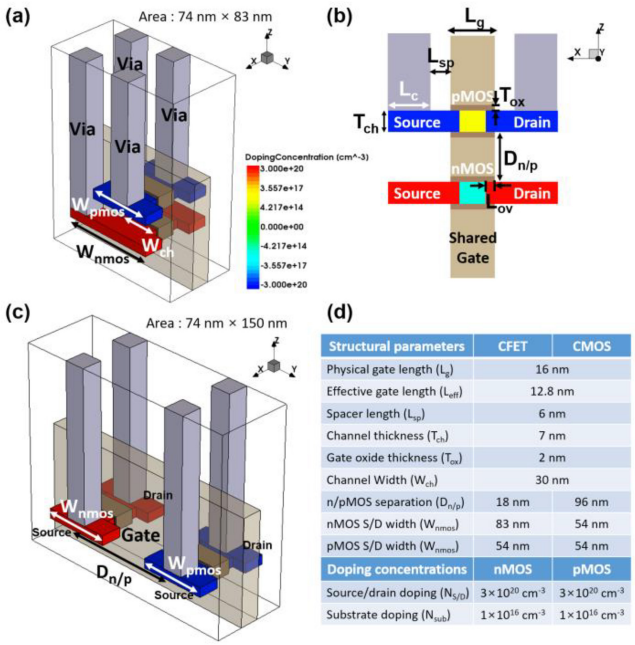
ABSTRACT For the first time, by using 3-D TCAD, the advantage of using complementary FET (CFET), which has vertically stacked nanosheet nFET and pFET with shared gate, is compared to standard CMOS with nanosheet FETs in perspective of CMOS inverter performance. The comparative study on CMOS operation was performed between CFET and standard CMOS in 3-nm technology node. The results indicate that, when both devices have identical DC electrical characteristics, using CFET can increase the frequency by ∼2.3% in iso-power and decrease power by ∼7.3% in iso-frequency compared to the standard CMOS with separate n/pFETs while effectively reducing the area by ∼55%. It is also investigated that such results are due to the approximately 4.5% low effective capacitance (C_{eff}) of the CFET compared to the standard CMOS. This low C_{eff} of CFET arises from the stacked structure, which causes the gate-fringe electric field overlap and short via pitch between nFET and pFET. Furthermore, the performance of CFET by different n/pFET separation distances, channel lengths, and widths are analyzed. This study can provide critical insight into the performance improvement by using CFET for sub 3-nm technology.

INDEX TERMS Complementary FET (CFET), Nanosheet FET (NSHFET), CMOS inverter, Technology computer-aided design (TCAD), 3-nm technology node.

I. INTRODUCTION

For next generation of logic devices after fin field effect transistors (FinFETs), gate-all-around (GAA) FETs, or nanosheet (NSH) FETs are considered promising candidates owing to their superior gate controllability and scalability compared to FinFETs [1], [2]. Currently, NSHFETs are actively developed in industry for production of logic devices under 5-nm technology node, but these devices still have physical limits in scaling down because of too delicate device fabrication process. Also, to develop the device under 3-nm technology node, more reduction in number of tracks and layout area is required [6], [7]. Thus, to further reduce the area of devices in layout, lots of device structures like forksheet FET, complementary field-effect transistor (CFET) have been suggested [1]–[7]. Among them, a CFET, which stacked n-type NSHFET and p-type NSHFET vertically with a shared gate for CMOS logic operation in single device, has recently been researched by Intel, IMEC and NARLab [3]–[5]. They demonstrate that CFET is one of the promising techniques beyond FinFET and NSHFET, which can effectively reduce a device area with high compatibility to conventional FinFET and NSHFET fabrication process [3]–[5]. Thus, there have been several researches on CMOS inverter operation of CFET, but they include buried power rail (BPR) technique which is unusual in industry [2], [8]. Except them, most of the researches on general CFET only focus on developing fabrication technique and do not deal with CMOS inverter operation specifically [3]–[5]. In case of NARLab, they predict that CFET can improve CMOS inverter performances when compared to standard CMOS with NSHFETs, but there is no analysis on the reason [5]. However, prediction and analysis of CMOS inverter performances is crucial to ascertain the possibility of commercialization [6], [7].

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Structural parameters	CFET	CMOS
Physical gate length (L_g)	16 nm	
Effective gate length (L_{eff})	12.8 nm	
Spacer length (L_{sp})	6 nm	
Channel thickness (T_{ch})	7 nm	
Gate oxide thickness (T_{ox})	2 nm	
Channel Width (W_{ch})	30 nm	
n/pMOS separation ($D_{n/p}$)	18 nm	96 nm
nMOS S/D width (W_{nmos})	83 nm	54 nm
pMOS S/D width (W_{nmos})	54 nm	54 nm
Doping concentrations	**nMOS**	**pMOS**
Source/drain doping ($N_{S/D}$)	3×10^{20} cm^{-3}	3×10^{20} cm^{-3}
Substrate doping (N_{sub})	1×10^{16} cm^{-3}	1×10^{16} cm^{-3}

FIGURE 1. **(a) 3-D schematic of CFET and (b) the CFET's cross sectional view through channel, (c) 3-D schematic of the standard CMOS with separate n/p nanosheet FETs, (d) structural specifications referenced from 3-nm technology node in IRDS 2020.**

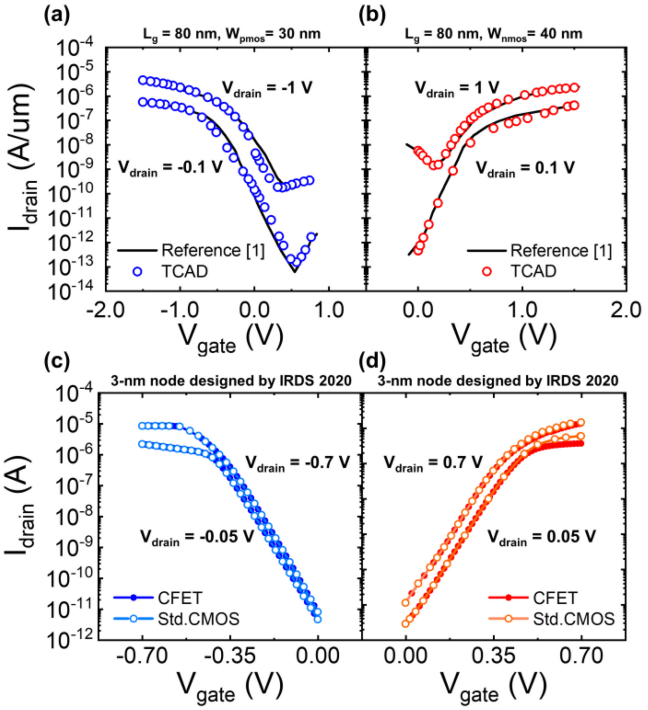


FIGURE 2. **Calibrated transfer characteristics of (a) pFET and (b) nFET of CFET between experimental [3] and TCAD simulation. Transfer characteristics of (c) pFET and (d) nFET of CFET and the standard CMOS designed for 3-nm technology node in IRDS 2020.**

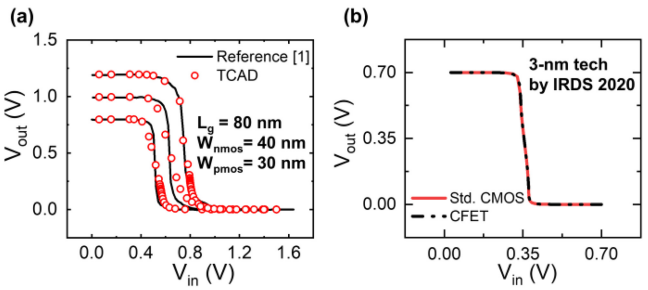


FIGURE 3. **(a) Calibrated voltage transfer characteristic (VTC) of CFET between experimental [3] and TCAD simulation in power supply voltage (V_{dd}) of 0.8V, 1.0V and 1.2V. (b) VTC of CFET and standard CMOS designed for 3-nm technology node in V_{dd} of 0.7V.**

For the first time, in perspective of CMOS operation, the advantages of using CFET compared to standard CMOS with NSHFETs is investigated by calibrated 3-D TCAD. A comparison of the effective capacitance (C_{eff}), effective resistance (R_{eff}), frequency (f), and power are carried out for CFET and the standard CMOS in 3-nm technology node, when both CFET and standard CMOS have the same DC inverter characteristics. Moreover, f and the power of CFET are investigated using different n/pFET separation distances ($D_{n/p}$), gate lengths (L_g), and channel widths (W_{ch}).

II. MODELING METHODOLOGY

The 3-nm CFET and the standard CMOS with separate n/pFETs are designed based on the IRDS 2020 high performance specification in Sentaurus 3-D TCAD vQ 2019. 12 [9]. The structure of CFET is referred to experimental [3]. Fig. 1 (a) and (c) show the 3-D bird's eye view of the CFET and the standard CMOS with separate n/pFETs, respectively. The stacked structure of nanosheet in CFET is fabricated from SiO_2/Si/SiO_2/Si structure as same as [3]. It is assumed that pFET and nFET are located in the upper and lower layers, respectively [3]. Physical gate length (L_g) is selected of 16 nm and Effective gate length (L_{eff}) is selected of 12.8 nm. 2-nm thickness of HfO_2 is used for gate oxide. In addition, TiN, which has a work function of 4.54 eV, is used as the common gate of CFET. Subsequently, SiO_2 is used for oxide passivation. Gate spacer and space between nFET and pFET are also filled with SiO_2. Then, W metal via is formed and the top side of via is selected of physical contact in TCAD. The standard CMOS structure is generated to have same nanosheet channel dimension to CFET. The device layout of standard CMOS is considered by a 1X inverter in [10]. Distance between center of nFET and pFET in standard CMOS is selected of 96 nm, which is sum of device width of 54 nm and space between nFET and pFET of 42 nm in IRDS 2020 [9]. The structural specifications used in the simulations are shown in Fig. 1 (d).

For the device physics, models of density gradient quantization and inversion accumulation mobility are used to consider carrier transportation in thin film with quantum confinement, and models of bandgap narrowing, electric field dependent mobility, doping-dependent mobility, high field saturation, and Shockley–Read–Hall doping dependence are used. To obtain accuracy, in Fig. 2 (a), (b) and Fig. 3 (a), the transfer characteristics and voltage transfer characteristic (VTC) of CFET was calibrated to CFET in

TABLE 1. Performance parameters of transfer characteristic of CFET and standard CMOS in 3-nm technology node.

Structure	Type	I_{off} [A]	I_{on} [A]	SS [mV/dec]
CFET	nMOS	1.10×10^{-10}	1.36×10^{-5}	78.73
	pMOS	7.87×10^{-12}	8.55×10^{-6}	78.68
Std. CMOS	nMOS	1.10×10^{-10}	1.48×10^{-5}	78.61
	pMOS	7.87×10^{-12}	8.55×10^{-6}	78.68

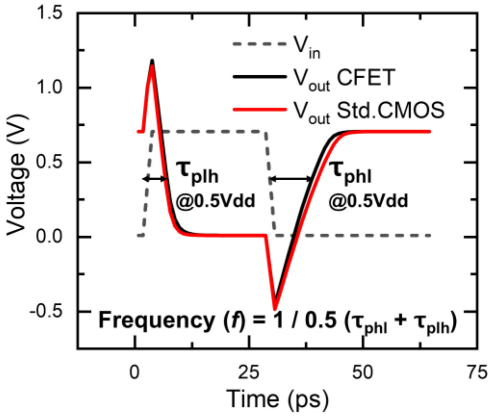


FIGURE 4. Transient response of CFET and standard CMOS designed for 3-nm technology node in ~17-GHz inverter operation.

gate length of 80 nm from experimental [3]. Then, 3-nm CFET was designed based on IRDS 2020, and DC and AC performances are calculated as shown in Fig. 2 (c), (d) and Fig. 3 (b). The VTC and transient response of the devices are calculated from "mixed-mode" in physics of SDEVICE of sentaurus TCAD. The default value of power supply voltage (V_{dd}) is set for 0.7 V. For transient response, fan-out of 3 (FO3) without load capacitance (C_{load}) is assumed [5]–[7]. C_{eff} is calculated from general equation of C_{eff} = total gate capacitance (C_{gg}) + total drain capacitance (C_{dd}) + 3 (number of fan-out devices) × fan-out capacitance (C_{fo}) [11], [12]. Here, C_{fo} is assumed as same to C_{gg}. The equation of C_{eff} is based on general miller capacitance theory in CMOS operation. In addition, the C_{gg} includes gate-to-drain capacitance (C_{gd}), gate-to-source of nFET (C_{gsn}) and gate-to-source of pFET (C_{gsp}). The C_{gg} and C_{dd} is extracted from single CFET device with "mixed-mode" in Sentaurus SDEVICE. Here, internal power is not considered as the portion of dynamic power by gate fringing capacitance dominates [9]. f is extracted by using the equation shown in Fig. 4. The power is calculated by $C_{eff} \times f \times V_{dd}^2$, and R_{eff} is calculated as $1/(C_{eff} \times f)$ [11]. Note that, all results are for front-end-of-line (FEOL).

III. RESULTS AND DISCUSSIONS

Fig. 2 shows transfer characteristics of CFET and standard CMOS. In Fig. 2 (a) and (b), transfer characteristics of the devices are calibrated in [3] and they matched well. Then, in Fig. 2 (c) and (d), transfer characteristics of CFET and standard CMOS designed for 3-nm technology are extracted. The performance parameters of transfer characteristic are shown in Table 1. Here, both CFET and standard CMOS shows

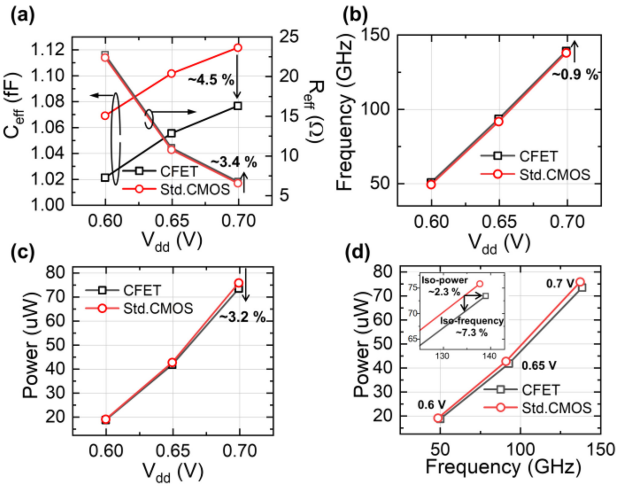


FIGURE 5. (a) C_{eff} and R_{eff} versus V_{dd}; (b) f versus V_{dd}; (c) power versus V_{dd}; and (d) power versus f of CFET and standard CMOS.

almost identical transfer characteristic except for some slight differences. The on-state current (I_{on}) of nFET of CFET was slightly lower compared to standard CMOS because of longer drain to source length. Thus, subthreshold swing (SS) was also slightly different between CFET (78.73 mV/dec) and standard CMOS (78.61 mV/dec). Fig. 3 shows the VTC of CFET and standard CMOS. In Fig. 3 (a), the VTC of the devices are calibrated in [3] and it matched well. Fig. 3 (b) shows the VTC of the CFET and the standard CMOS designed for 3-nm technology. The VTC between the two devices show identical values. The maximum gain of VTC is ~10.4 V/V at V_{in} = 0.35 V. Fig. 4 shows the transient response of single CFET and the standard CMOS in a ~17-GHz inverter operation. Here, the f of the device is calculated by the inverse of the average propagation delay time from high to low (T_{phl}) and propagation delay time from low to high (T_{plh}) at 0.5 V_{dd} [11]. Subsequently, C_{eff}, R_{eff}, f, and power are investigated.

In Fig. 5, C_{eff}, R_{eff}, f, power of CFET, and standard CMOS by different V_{dd} are compared and analyzed. In Fig. 5 (a), C_{eff} and R_{eff} of the devices by different V_{dd} from 0.6 V to 0.7 V are investigated. Here, the C_{eff} of CFET is ~1.076 fF which is ~4.5% lower than the standard CMOS for ~1.126 fF at V_{dd} of 0.7 V and the tendency is maintained for V_{dd} of 0.6 V. In the case of R_{eff} in Fig. 5 (a), R_{eff} of CFET is ~3.4 % higher than the standard CMOS at V_{dd} = 0.7 V and the tendency is maintained for V_{dd} of 0.6 V. For f in Fig. 5 (b), f of CFET is ~0.9% higher than that of the standard CMOS at V_{dd} = 0.7 V and the tendency is maintained for V_{dd} of 0.6 V. Here, CFET shows higher f than standard CMOS although CFET has high R_{eff} compared to standard CMOS. The f is proportional to $1/(R_{eff} \times C_{eff})$ [11]. The R_{eff} of CFET is ~3.4% higher than standard CMOS, as nFET of CFET shows slightly lower on-state current compared to standard CMOS (Fig. 2 (d)). However, ~4.5% lower C_{eff} of CFET overcomes the degradation of f by high R_{eff} of CFET. With regard to the power of Fig. 5 (c), CFET can save power for ~3.2% compared to the standard CMOS at V_{dd} = 0.7 V and the tendency is maintained for V_{dd} of

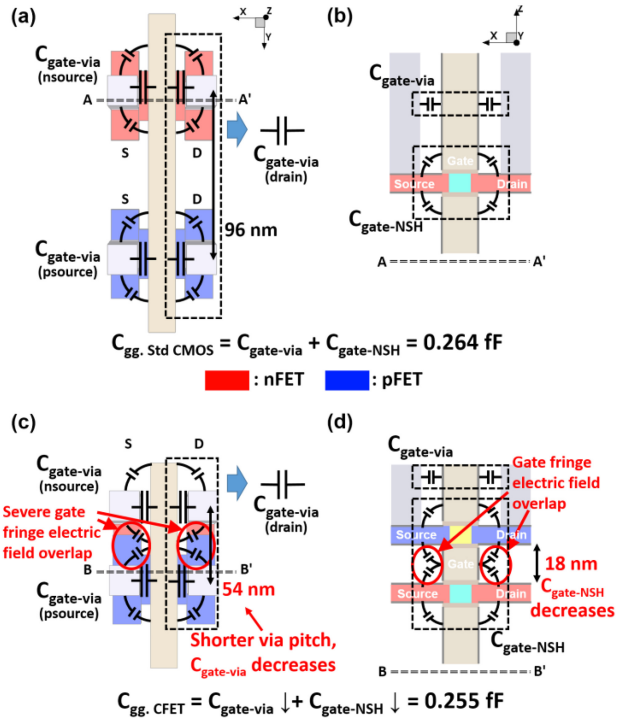


FIGURE 6. **(a) Top view and (b) cross-sectional view of standard CMOS structure with schematic of capacitive component. (c) Top view and (d) cross-sectional view of default CFET structure with schematic of capacitive component.**

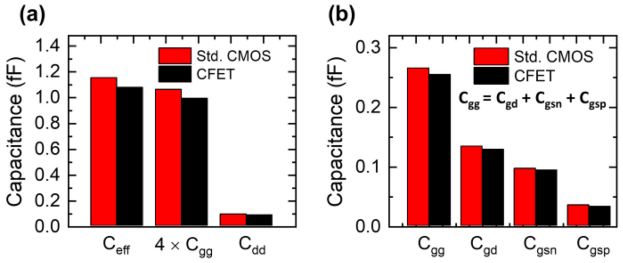


FIGURE 7. **Capacitance of CFET and standard CMOS extracted from mixed-mode TCAD when V_{dd} is 0.7 V. (a) Effective capacitance (C_{eff}) and its components of total gate capacitance (C_{gg}) and total drain capacitance (C_{dd}). (b) C_{gg} and its components of gate-to-drain capacitance (C_{gd}), gate-to-source of nFET (C_{gsn}) and gate-to-source of pFET (C_{gsp}).**

0.6 V. The Fig. 5 (d) shows power versus f graph of CFET and the standard CMOS. Here, the f-power graph indicates how f is different at iso-power and how power is different at iso-f between different device structure [6], [7]. In Fig. 5 (d), the CFET has a ∼2.3% higher f in iso-power of ∼70.2 μW, and ∼7.3% lower power in iso-f of ∼134.6 GHz compared to the standard CMOS.

C_{eff} is the most important factor that determines f and power [6], [7]. Especially, C_{gg} is the major components which determines the C_{eff} as $C_{eff} = \sim 4 \times C_{gg} + C_{dd}$ in FO3 structure. Fig. 6 illustrates the top and cross-sectional views of the standard CMOS and CFET with a schematic of the capacitive components of C_{gg}. The figures explain why CFET can reduce C_{gg} compared to standard CMOS. Here, C_{gg} can be simply distinguished by the capacitance between the metal gate and n/pFET source/drain metal via ($C_{gate\text{-}via}$), and the capacitance between the metal gate and the Si nanosheet ($C_{gate\text{-}NSH}$). Fig. 6 (a) and (c) explain difference of $C_{gate\text{-}via}$ between standard CMOS and CFET, respectively. The source/drain metal via pitch between n/pFET is 96 nm (Fig. 6 (a)), which is higher than that of the CFET, whose pitch is 54 nm (Fig. 6 (c)). The reduction of metal via pitch between nFET and pFET triggers decreasing of fringing capacitance (red circle in Fig. 6 (c)) of $C_{gate\text{-}via}$. As a result, the $C_{gate\text{-}via}$ in CFET is much lower than standard CMOS. For $C_{gate\text{-}NSH}$, in Fig. 6 (b) for the standard CMOS, a fringe electric field occurs between the gate and the nanosheet without any interference. However, in Fig. 6 (d) of CFET, stacking n/pFET triggers an overlap of the gate-fringe electric field between nFET and pFET. This triggers reduction of $C_{gate\text{-}NSH}$. As a result, reduction of both $C_{gate\text{-}via}$ and $C_{gate\text{-}NSH}$ decreases C_{gg} [3]–[5]. Fig. 7 shows capacitance components of CFET and standard CMOS. In Fig. 7 (a), C_{eff} and its components of $4 \times C_{gg}$ and C_{dd} are shown. For both CFET and standard CMOS, $4 \times C_{gg}$ shows ∼12 times higher value compared to C_{dd}. These emphasize that the analysis on C_{gg} is the most important factor in C_{eff}. The Fig. 7 (b) shows the C_{gg} and its components such as C_{gd}, C_{gsn} and C_{gsp}. For both CFET and standard CMOS, C_{gd} shows the highest value compared to C_{gsn} and C_{gsp}. This is because drain capacitance of nFET and pFET are considered as one, since drains of nFET and pFET are connected and have same voltage. Also, C_{gsn} shows ∼3 times larger value compared to C_{gsp} because of bottom nMOS's extended via for via landing. This occurs nMOS's via to be taller than pMOS's via. Comparing the capacitance between CFET and standard CMOS, all C_{gg}, C_{gd}, C_{gsn}, and C_{gsp} of CFET show lower values than that of standard CMOS. The C_{gg} of CFET shows 10.7 aF lower value compared to standard CMOS. For C_{gd}, C_{gsn} and C_{gsp} of CFET, they show 5.35, 2.79, 2.55 aF lower value respectively, compared to standard nMOS.

In Fig. 8, the power versus f graph, C_{eff}, and R_{eff} of CFET for different $D_{n/p}$, L_g, and W_{ch} are extracted to analyze the performance of the CFET in different dimensions. Fig. 8 (a) shows the f versus power graph for different $D_{n/p}$ values ranging from 18 nm to 200 nm. For the same V_{dd}, both f and power increase with a decrease in $D_{n/p}$. In addition, f increases in iso-power and power decreases in iso-f with reduction of $D_{n/p}$. Fig. 8 (b) shows the C_{eff} and R_{eff} of the CFET by $D_{n/p}$, and they decrease with a decrease in $D_{n/p}$. Here, f is proportional to $1/(R_{eff} \times C_{eff})$, and the power is calculated for $C_{eff} \times f \times V_{dd}^2$, which is proportional to $1/R_{eff}$. Thus, for the same V_{dd}, low R_{eff} and C_{eff} trigger higher f and power with a decrement in $D_{n/p}$, as shown in Fig. 8 (a). Furthermore, f is affected by both R_{eff} and C_{eff}, but the power is only affected by R_{eff}. Thus, the increment in f is much higher than the increment of power with L_g decrement. Therefore, as both R_{eff} and C_{eff} decreases with reduction of $D_{n/p}$, the f increases in iso-power and power decreases in iso-f with $D_{n/p}$ of reduction. Fig. 8 (c) shows the f-power graph for different values of L_g. Here, both f and power increase with a shorter L_g in the same V_{dd}. In addition, f

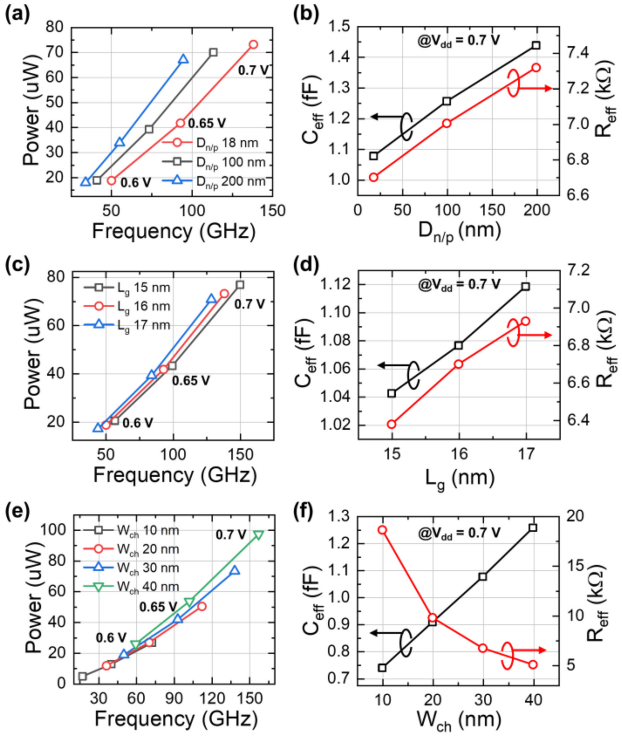


FIGURE 8. Power versus f graph of CFET by different (a) $D_{n/p}$, (c) L_g, (e) W_{ch} when V_{dd} is 0.6 V, 0.65 V, and 0.7 V. C_{eff} and R_{eff} of CFET versus different (b) $D_{n/p}$, (d) L_g, (f) W_{ch} of CFET when V_{dd} is 0.7 V.

increases in iso-power and power decreases in iso-f with reduction of L_g. Fig. 8 (d) shows the C_{eff} and R_{eff} of CFET for different L_g values, and both C_{eff} and R_{eff} decrease with a reduction in L_g. This is because a short L_g decreases the area of the gate and then decreases C_{eff}. In addition, a short L_g increases the drain current, thereby reducing R_{eff}. Thus, a low L_g reduces R_{eff} and C_{eff} and results in a higher f and power at the same V_{dd}. Furthermore, f is affected by both R_{eff} and C_{eff}, but the power is only affected by R_{eff}, so the increment in f is much higher than the increment of power with L_g decrement. Therefore, the f increases in iso-power and power decreases in iso-f with a decrease in L_g. Fig. 8 (e) shows the f-power graph for different values of W_{ch}. Both f and power decrease with a decrease in W_{ch} for the same V_{dd}. In addition, the f-power slope decreases with a decrease in W_{ch}. In Fig. 8 (f), C_{eff} decreases, but R_{eff} increases with the reduction in W_{ch}. A short W_{ch} decreases the area of the gate oxide and capacitance of the gate oxide, and it decreases C_{eff}. In addition, a short W_{ch} decreases the current, thereby increasing R_{eff}. Although C_{eff} decreases, much higher R_{eff} makes f becomes lower in same V_{dd} with decrement of W_{ch}. The power decreases in same V_{dd} with decrement of W_{ch} as power is proportional to $1/R_{eff}$ and R_{eff} increases. Furthermore, as decrement of power is much higher than f with decrement of W_{ch}, the f increases in iso-power and power decreases in iso-f with a decrease in W_{ch}.

IV. CONCLUSION

In this paper, the merits in performance of CFET compared to standard CMOS with NSHFETs is firstly investigated in perspective of f and power using 3-D TCAD. The C_{eff}, R_{eff}, f, and power between CFET and the standard CMOS are compared for same design specification of 3-nm technology node. The results show that the CFET has 0.9% higher f and 3.2% lower power for the same V_{dd} of 0.7 V compared to the standard CMOS, when CFET and standard CMOS have same DC inverter characteristics. In addition, CFET shows ~2.3% higher f in iso-power and ~7.3% lower power in iso-f compared to the standard CMOS while effectively reducing the area by ~55%. These performance enhancements of CFET compared to standard CMOS are caused by low C_{eff} of CFET. The CFET shows ~4.5% lower C_{eff} compared to the standard CMOS owing to the gate-fringe electric field overlap and short via pitch between n/pFET, which are triggered by the stacked n/pFET structure of CFET. In addition, C_{eff}, R_{eff}, f, and power by different $D_{n/p}$, L_g, and W_{ch} are investigated. For $D_{n/p}$ of CFET, a short $D_{n/p}$ decreases C_{eff} owing to the gate-fringe electric field overlap. For L_g and W_{ch}, C_{eff} decreased as the area of the gate oxide reduced. Therefore, when $D_{n/p}$, L_g, and W_{ch} decrease, f increases in iso-power, and power decreases in iso-f. This work provides insight that gate-fringe electric field overlap is crucial for performance improvement when using CFET for sub-3 nm CMOS technology in perspective of FEOL.

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